Query-efficient distance amplification for non-adaptive locally decodable codes

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Editorial note. Faithful editable transcription of the official CCC2021 CC BY4.0 publication, article1 pp1:1–1:57. This item collects the non-explicit distance-amplification main result Theorem72, complete Section6.1 construction and Section6.2 analysis including Proposition65 and full Claims66–71 proofs, Section6.2.1 complete proof and low-error/direct-instantiation discussion, Definition64, and supporting sampler Definition14/Theorem15 full author proof sketch/Lemma16 plus standard Theorem18. Theorem73 is retained as an original cited prerequisite and direct instantiation, not counted separately. All43 original bibliography entries retained. Original wording and indices, including Claim68 k3 and Claim69 v notin Bz, are preserved without mathematical correction. Author nonadaptivity, alphabets, constant rate/error, sampler degree conventions and all query/error dependencies retained. Explicit variants Sections6.3–6.6 are outside this item; the complete official original is preserved. No native TeX was found in the bounded source lookup; this is a transcription of the licensed official PDF. Root handles compilation and database delivery.

Allowed standard prerequisites (editorial).

- Chernoff bound and binary entropy bound. Original sampler proof sketch and Lemma16 retained; no recursion into standard probability proofs.
- Gilbert-Varshamov codes. Original Theorem18 statement retained.
- Asymptotically good locally decodable codes. Original Theorem73[26] statement retained as optional direct instantiation.

Theorem 72 (Author Theorem 72)**.** *Let C be a block-length-n $(q, \delta, 1/5)$-LDC over alphabet Σ having a constant rate. Let C' be a family of asymptotically good $(q'_n, \delta', 1/5)$-LDC, where q'_n is the query complexity when the code from the family is taken with block length n. Then, there exists an asymptotically good LDC over Σ, with constant error, having block length $\Theta(n)$ and query complexity*

$$q_{\text{new}} = O\left(q \cdot q'_{O(1/\delta)} \log(1/\delta) \log n\right). \tag{6.2}$$

Author background and Section 3.1: Samplers

Recall that a function $C : \Sigma^k \to \Sigma^n$ is an error-correcting code with distance δ if for every distinct $x, y \in \Sigma^k$, $\mathrm{dist}(C(x), C(y)) \geq \delta$, where dist is the relative Hamming distance. The rate of the code C is given by $\rho = k/n$. When there is more than one possible message length, we consider a code family, which is a family of functions in which each function is a code, and there is one code per message length k. A code family is asymptotically good if both the rate and distance of every code in the family are uniformly bounded below by constants $\rho > 0$ and $\delta > 0$, respectively. Our distance amplification procedure makes use of samplers. These are bipartite graphs with a certain pseudo-random property. Let $G = (L, R, E)$ be a bipartite graph. We say G is left-regular if all nodes in L have the same degree.

Definition 14 (Author Definition 14, [9])**.** *Let $0 < \varepsilon, \delta < 1$. A bipartite graph $G = (L, R, E)$ is an (ε, δ)-sampler if for every subset $T \subseteq R$, for all but δ-fraction of vertices $v \in L$ it holds that*

$$\left| \frac{|\Gamma(v) \cap T|}{|\Gamma(v)|} - \mu(T) \right| \leq \varepsilon.$$

We will be working with "unbalanced" samplers. These are samplers with $|L| \gg |R|$. The state-of-the-art constructions of these samplers rely on their connection to randomness seeded extractors. We refer the interested reader to the excellent survey by Goldreich [17] for more information. When working with samplers, it is rather typical that the bipartite graph is left-regular, that is, the degree of all vertices in L is the same. A small additional technical property we need is that the degree of every vertex in R is close to the average right-degree. We make use of the following theorem which gives (non-explicit) samplers with near-optimal parameters having the above properties with respect to the degrees. We give a proof sketch for completeness.

Theorem 15 (Author Theorem 15)**.** *There exists a universal constant $c_{\mathrm{samp}} \geq 1$ such that the following holds. For all integers ℓ, r and all $\varepsilon > 0$, $1/2 > \delta > 0$ for which $\ell \geq r/(\delta \log(1/\delta))$, there exists a left-regular (ε, δ)-sampler $G = ([\ell], [r], E)$ with left-degree $d = c_{\mathrm{samp}} \cdot \log(1/\delta)/\varepsilon^2$. Moreover, provided that $\log r < 1/(\delta\varepsilon^2)$, every right vertex has degree in $[D/2, 2D]$ where $D = \ell d/r$ is the average right degree.*

For the proof we need the following well-known lemma.

Lemma 16 (Author Lemma 16)**.** *For every integers $1 \leq k \leq n$ with $k/n = \delta \leq 1/2$ it holds that*

$$\sum_{i=0}^{k} \binom{n}{i} \leq 2^{H(\delta)n},$$

where $H(x) = -x \log(x) - (1-x)\log(1-x)$ is the binary entropy function.

Author proof sketch for Theorem 15. The proof is via the probabilistic method, where for every left vertex we choose d neighbours independently and uniformly at random, and independently across all left vertices (note that in the above we allow for parallel edges, but if that troubles the reader, that can be avoided

as well in the regime of interest $d \ll r$ by arguing that the probability of a right neighbor to be selected more than once is small. In any case, our distance amplification procedure works just as well with parallel edges). Fix $T \subseteq [r]$. For $v \in [\ell]$ let F_v be the indicator random variables that is 1 if and only if $\left||\Gamma(v) \cap T|/d - \mu(T)\right| > \varepsilon$. By the Chernoff bound, $\Pr[F_v] \leq e^{-\Omega(\varepsilon^2 d)}$. Fix $S \subseteq [\ell]$ with $|S| = \delta\ell$. The probability that for all vertices $v \in S$ it holds that $F_v = 1$ is bounded above by $e^{-\Omega(\varepsilon^2 d \cdot \delta\ell)}$. By taking the union bound over all S and T, we get that except with probability

$$2^r \binom{\ell}{\delta\ell} e^{-\Omega(\varepsilon^2 d\delta\ell)} \leq 2^{r + H(\delta)\ell - c\varepsilon^2 d\delta\ell}, \tag{3.1}$$

the sampled graph is an (ε, δ)-sampler. Note that the last inequality follows by Lemma 16, where $c > 0$ is some constant. By taking $c_{\text{samp}} \geq 5/c$, one can verify (using that $H(x) \leq 2x \log(1/x)$ for all $x \leq 1/2$) that the right hand side in Equation (3.1) is bounded by $1/4$. As for the moreover part, again, by the Chernoff bound, the probability that there exists a right vertex which has degree outside $[D/2, 2D]$ is bounded above by $re^{-\Omega(\ell d/r)}$, and this is bounded by $1/4$ by our choice of parameters and by taking c_{samp} large enough. □

Theorem 18 (Author Theorem 18: The Gilbert–Varshamov bound)**.** *Let Σ be a set of size $|\Sigma| = q$. For every $n \in \mathbb{N}$, and $0 \leq \delta \leq 1 - 1/q$ there exists a code of block-length n over Σ, with distance at least δ and rate $r \geq 1 - H_q(\delta)$. Furthermore, if q is a prime power and $\Sigma = \mathbb{F}_q$, there exists a linear code over Σ with rate $r \geq 1 - H_q(\delta) - g(n)$, where $g(n) = O(1/n)$.*

6 Query-efficient distance amplification

In this section we construct our query-efficient distance amplification procedure. We start by giving a somewhat more formal definition of locally decodable codes (compared to Definition 1) or, more precisely, a more formal definition of their non-adaptive counterparts. Recall that, informally, these are LDC in which the joint distribution of queries depends solely on the index one wishes to decode and is independent of the received word. By inspection, it is our understanding that the AEL distance amplification procedure also requires non-adaptivity.

Definition 64 (Author Definition 64: Locally decodable codes)**.** *Let (C, Q, R) be a tuple of functions*

$$\begin{aligned} C &: \Sigma_{\text{in}}^k \to \Sigma_{\text{out}}^n, \\ Q &: [k] \times \{0,1\}^r \to [n]^q, \\ R &: [k] \times \Sigma_{\text{out}}^q \times \{0,1\}^r \to \Sigma_{\text{in}}. \end{aligned}$$

Define $D : [k] \times \Sigma_{\text{out}}^n \times \{0,1\}^r \to \Sigma_{\text{in}}$ as follows. For $v \in [k]$, $y \in \Sigma_{\text{out}}^n$, and $s \in \{0,1\}^r$, let

$$Q(v, s) = (u_1, \ldots, u_q), \qquad D(v, y, s) = R(v, y_{u_1}, \ldots, y_{u_q}, s).$$

The tuple (C, Q, R) is called a (q, δ, ε)-locally decodable code (or (q, δ, ε)-LDC for short) if the following holds. For every $v \in [k]$, $x \in \Sigma_{\text{in}}^k$, and $y \in \Sigma_{\text{out}}^n$ for

which $\mathrm{dist}(y, C(x)) \leq \delta$*, it holds that*

$$\Pr_{s \sim U_r}[D(v, y, s) = x_v] \geq 1 - \varepsilon.$$

We call the function C the encoding function, Q the querying function, and R the reconstruction function. The induced function D is called the decoding function. The parameters k, n are referred to as the message length and the block length, respectively. The sets $\Sigma_{\mathrm{in}}, \Sigma_{\mathrm{out}}$ are called the input alphabet and output alphabet, respectively. We will be interested mostly in locally decodable codes in which $\Sigma_{\mathrm{in}} = \Sigma_{\mathrm{out}}$ in which case we refer to both as the alphabet of the code. The parameter r is called the randomness complexity of the LDC. We say the LDC is explicit if all three functions C, Q, R are polynomial-time computable. Note that then the decoding function D is also polynomial-time computable.

6.1 The distance amplification procedure

In this section we present our query-efficient distance amplification procedure. We start by describing the building blocks we use and specify their parameters.

Building blocks

- For $i = 1, 2$ let (C_i, Q_i, R_i) be a $(q_i, \delta_i, \varepsilon_i)$-LDC with message length k_i and block length n_i over the same alphabet Σ. We denote the rate k_i/n_i of C_i by ρ_i.
- Let (C_3, Q_3, R_3) be a family of $(q_3(k_3), \delta_3(k_3), \varepsilon_3(k_3))$-LDC having rate $\rho_3(k_3)$ for message length k_3. The code C_3 is also over the alphabet Σ. We will always work with functions $q_3, \delta_3, \varepsilon_3, \rho_3$ that are monotone. More precisely, q_3 and ρ_3 are non-decreasing and δ_3, ε_3 are non-increasing. We sometimes write $q_3, \delta_3, \varepsilon_3, \rho_3$ without mentioning explicitly the message length, and by that refer to the largest k_3 considered in the construction for q_3, δ_3 and the smallest k_3 when considering ε_3, ρ_3. In any case, we assume (mostly for simplicity) that $\rho_3(k_3) \geq 1/2$ for all k_3.
- Set $\ell = n_1/k_2$. Let $G = (L, R, E)$ be a $(\delta_2/2, \delta_1)$-sampler with $|L| = \ell$ and $|R| = r$. Assume G is left-regular with left-degree $d = n_2$. Assume further that every right-vertex v of G has degree $\deg(v) \in [D/2, 2D]$, where D is the average right degree $D = \ell d/r = n_1/(r\rho_2)$.

How to think of the parameters? We think of C_1 as the code whose distance δ_1 we wish to amplify. Typically, the code C_2 has a much shorter message length $n_2 \ll n_1$. In all applications in this paper we take δ_2 to be either constant or slightly sub constant in n_1. The code C_3 has a larger block length than C_2 and, depending on the application, it has either a somewhat smaller or much smaller message length than n_1. We typically take $\delta_3 \approx \delta_2$. The rates of all three codes is taken to be constant and even close to one. Note that we take C_3 to be a family of codes, whereas C_1 and C_2 are codes with predetermined message lengths. The reason is that the sampler G is not necessarily right-regular, and in the construction, we associate codes from C_3 with the right vertices of G. Recall, though that the ratio of largest to smallest right-degree is bounded by 4, so that is a minor technicality. To describe the

LDC that is composed of these building blocks, we need to specify the encoding function, querying function and reconstruction function. We start by describing the encoding function.

The encoding function Let $n = \sum_{v \in R} n_v$ where n_v is the block length of the code from the family C_3 having message length $k_v = \deg(v)$. We define the function $C : \Sigma^{k_1} \to \Sigma^n$ as follows. Let $x \in \Sigma^{k_1}$.

1. Compute $y = C_1(x) \in \Sigma^{n_1}$.
2. Partition y to $y = y^{(1)} \circ \cdots \circ y^{(\ell)}$ consecutive blocks, each of length k_2. Recall that, indeed, $n_1 = \ell k_2$.
3. For every $u \in [\ell]$ compute $z^{(u)} = C_2(y^{(u)}) \in \Sigma^{n_2}$.
4. For every $v \in [r]$ and $j \in [\deg(v)]$ let $(u, j') = \Gamma(v, j) \in [\ell] \times [n_2]$. Define the string $w^{(v)} \in \Sigma^{\deg(v)} = \Sigma^{k_v}$ as follows: for $j \in [\deg(v)]$, $(w^{(v)})_j = (z^{(u)})_{j'}$.
5. For every $v \in [r]$ compute $t^{(v)} = C_3(w^{(v)}) \in \Sigma^{n_v}$.
6. The output of the encoding function on input x is then defined by $C(x) = t^{(1)} \circ \cdots \circ t^{(r)} \in \Sigma^n$, where as usual we identify R with $[r]$.

By the construction of the encoding function, the message length and block length of the resulted code are k_1 and n, respectively. From here on we denote $k = k_1$.

The querying function We denote the randomness complexity of C_1, C_2, C_3 by r_1, r_2, r_3, respectively. The randomness complexity of the resulting querying function will be $r = r_1 + r_2 + r_3$, and the query complexity will be $q \leq q_1 q_2 q_3$, where q_3 is taken to be the maximum query complexity taken over all right vertices. We turn to define the querying function $Q : [k] \times \{0,1\}^r \to [n]^q$ as follows. On inputs $p \in [k]$, $s \in \{0,1\}^r$ we proceed as follows.

1. Partition $s = s_1 \circ s_2 \circ s_3$ where $|s_1| = r_1$, $|s_2| = r_2$, $|s_3| = r_3$.
2. Compute $(a_1, \ldots, a_{q_1}) = Q_1(p, s_1) \in [n_1]^{q_1}$.
3. For $i = 1, \ldots, q_1$
   (a) Set $u_i = \lceil a_i / k_2 \rceil$ and $b_i = 1 + ((a_i - 1) \bmod k_2)$. Informally, u_i is the "bucket" in which a_i resides and b_i is its location within the bucket. Note that we start the counting from 1 rather than 0, hence the slightly annoying addition and subtraction by 1 in the definition of b_i.
   (b) Compute $(t_1^{(i)}, \ldots, t_{q_2}^{(i)}) = Q_2(b_i, s_2) \in [n_2]^{q_2}$.
   (c) For $j = 1, \ldots, q_2$
      i. Let $(v^{(i,j)}, \widehat{t}_j^{(i)}) = \Gamma(u_i, t_j^{(i)}) \in [r] \times [k_{v^{(i,j)}}]$.
      ii. Compute $(e_1^{(i,j)}, \ldots, e_{q_3}^{(i,j)}) = Q_3(\widehat{t}_j^{(i)}, s_3) \in [n_{v^{(i,j)}}]^{q_3}$.

iii. As before, we endow the right vertices of the sampler in a fixed (arbitrary) order by identifying R with $[r]$. For $h = 1, \ldots, q_3$ set $c^{(i,j,h)}$ to be the absolute location of $e_h^{(i,j)}$ in the ordering of R. That is, $c^{(i,j,h)} = e_h^{(i,j)} + \sum_{v<v^{(i,j)}} n_v$.

4. The result is then given by $Q(p,s) = (c^{(i,j,h)})_{(i,j,h)\in[q_1]\times[q_2]\times[q_3]}$.

Note that, indeed, the query complexity q of the querying function defined above is at most $q_1q_2q_3$ where, recall, $q_3 = q_3(2D)$. From here on we identify $[q]$ with $[q_1] \times [q_2] \times [q_3]$.

The reconstruction procedure We define the reconstruction procedure $R : [k]\times\Sigma^q\times\{0,1\}^r \to \Sigma$ as follows. On inputs $p \in [k]$, $\sigma = (\sigma^{(i,j,h)})_{(i,j,h)\in[q_1]\times[q_2]\times[q_3]} \in \Sigma^q$, and $s \in \{0,1\}^r$, we proceed as follows.

1. Partition $s = s_1 \circ s_2 \circ s_3$ where $|s_1| = r_1$, $|s_2| = r_2$, $|s_3| = r_3$ as in the querying function.
2. For $i = 1, \ldots, q_1$
   (a) For $j = 1, \ldots, q_2$
      i. Denote $(z_1, \ldots, z_{q_3}) = (\sigma^{(i,j,1)}, \ldots, \sigma^{(i,j,q_3)})$.
      ii. Compute $y_j^{(i)} = R_3(\widehat{t}_j^{(i)}, z_1, \ldots, z_{q_3}, s_3)$, where $\widehat{t}_j^{(i)} = \widehat{t}_j^{(i)}(p,s)$ as defined in the querying function.
   (b) Set $x_i = R_2(b_i, y_1^{(i)}, \ldots, y_{q_2}^{(i)}, s_2)$ where $b_i = b_i(p,s)$ as defined in the querying function.
3. The output is then given by $R(p,\sigma,s) = R_1(p, x_1, \ldots, x_{q_1}, s_1)$.

6.2 Analysis

In this section we analyze the LDC obtained above. We prove

Proposition 65 (Author Proposition 65)**.** *With the notation of the previous section, C is a (q,δ,ε)-LDC, where*

$$q \le q_1q_2q_3, \qquad \delta \ge \frac{\delta_2\delta_3}{16}, \qquad \varepsilon \le \varepsilon_1 + (\varepsilon_2+\varepsilon_3)n.$$

Furthermore, C has rate $\rho_1\rho_2\rho_3$, where ρ_1, ρ_2 are as defined in the building blocks paragraph, and per our convention set above, $\rho_3 = \rho_3(D/2)$.

Remark regarding the distance Note that the distance δ of the resulted code C is independent of δ_1, the poor distance of C_1 we set out to amplify. This is the key feature of the AEL distance amplification procedure (which our variant above, of course, maintains). It is yet another instance of a general strategy in pseudo-randomness that combines objects in such a way that the resulted object enjoys the upsides of the different parts and avoid their shortcomings. The Zig-Zag product is another classic example. But, of course, δ_1 has some effect on the resulted code. The effect δ_1 has on the code is via the query complexity. Indeed, as the analysis will show, the smaller δ_1 is (i.e., the weaker the guarantee

we have on the distance of C_1), the larger $k_2 = k_2(\delta_1)$ and $k_3 = k_3(\delta_1)$ must be, with a far stronger effect on k_3. More quantitatively, roughly speaking, by taking a sufficiently good sampler (e.g., the one that is given by Theorem 15), $k_2 \approx \operatorname{poly}\log(1/\delta_1)$ and $k_3 \approx \operatorname{poly}(1/\delta)$. This, in turn, effects the query complexities $q_2 = q_2(k_2)$ and $q_3 = q_3(k_3)$.

Author proof of Proposition 65. That the query complexity is $q \leq q_1 q_2 q_3$ readily follows by the querying function, where recall that per our convention $q_3 = q_3(2D)$. To analyze the rate, recall that ρ_3 is a non-decreasing function. Further, our convention dictates that by writing ρ_3 without explicitly mentioning the message length, we refer to ρ_3 applied with the smallest message length taken by the construction, namely, $\rho_3 = \rho_3(D/2)$. Thus,

$$n = \sum_{v \in R} n_v = \sum_{v \in R} \frac{k_v}{\rho_3(k_v)} \leq \frac{1}{\rho_3} \sum_{v \in R} k_v = \frac{\ell n_2}{\rho_3} = \frac{n_1}{\rho_2 \rho_3}.$$

Recall that $k = k_1 = \rho_1 n_1$ which shows that $\rho = k/n \geq \rho_1 \rho_2 \rho_3$. We turn to analyze the distance δ and error ε. Let $x \in \Sigma^k$ and let $\widetilde{C(x)} \in \Sigma^n$ be such that $\operatorname{dist}(C(x), \widetilde{C(x)}) \leq \delta$. Define the set of "errors", namely, the disagreements between $C(x)$ and $\widetilde{C(x)}$ by

$$B = \{i \in [n] \mid \widetilde{C(x)}_i \neq C(x)_i\}.$$

By assumption, $\mu(B) \leq \delta$. The error set B induces errors "backwards" throughout the construction. We proceed by analyzing these induced errors. Recall that, in the encoding function, we defined for each $v \in [r]$ an element $t^{(v)} = t^{(v)}(x) \in \Sigma^{n_v}$. Partition $\widetilde{C(x)}$ to r substrings $\widetilde{C(x)} = \widetilde{t}^{(1)} \circ \cdots \circ \widetilde{t}^{(r)}$, where $\widetilde{t}^{(v)}$ has length n_v, and define the set

$$B_t = \{v \in [r] \mid \operatorname{dist}(t^{(v)}, \widetilde{t}^{(v)}) \geq \delta_3\}.$$

Informally, $v \in B_t$ if the adversary has introduced too many errors on the respective block to allow for correct decoding via D_3.

Claim 66 (Author Claim 66)**.** $\mu(B_t) \leq 8\delta/\delta_3$.

Proof. For $v \in R$ let $e_v = \operatorname{dist}(t^{(v)}, \widetilde{t}^{(v)})$. We have that $\sum_{v \in R} e_v n_v \leq \delta n$. On the other hand,

$$\sum_{v \in R} e_v n_v \geq \delta_3 \sum_{v \in B_t} n_v \geq \frac{\delta_3 D |B_t|}{2}.$$

But, per our assumption that $\rho_3 \geq 1/2$, and since $k_v \leq 2D$ for all $v \in R$,

$$n = \sum_{v \in R} n_v \leq 2 \sum_{v \in R} k_v \leq 4Dr.$$

The proof follows by the above three inequalities. □

For convenience we also denote $B_w = B_t$. Next, we define

$$B_z = \{u \in [\ell] \mid |\Gamma(u) \cap B_w| \geq \delta_2 n_2\}. \tag{6.1}$$

Claim 67 (Author Claim 67)**.** $\mu(B_z) \leq \delta_1$.

Proof. By Claim 66 and by our assumption on δ,

$$\mu(B_w) \leq \frac{8\delta}{\delta_3} = \frac{\delta_2}{2}.$$

Recall that G is a $(\delta_2/2, \delta_1)$-sampler. Thus, at most δ_1-fraction of the left vertices $u \in [\ell]$ satisfy $\mu(\Gamma(u) \cap B_w) \geq \mu(B_w) + \delta_2/2$. The proof then follows since $\mu(B_w) \leq \delta_2/2$. □

Lastly, define

$$B_y = \left\{ a \in [n_1] \;\middle|\; \left\lceil \frac{a}{k_2} \right\rceil \in B_z \right\}.$$

For $v \in [r]$, $b \in [k_v]$ we define the function $\widetilde{w}_b^{(v)} : \{0,1\}^{r_3} \to \Sigma$ as follows: on input $s_3 \in \{0,1\}^{r_3}$

$$\widetilde{w}_b^{(v)}(s_3) = D_3(b, \widetilde{t}^{(v)}, s_3).$$

Claim 68 (Author Claim 68)**.** *There exists a set $E_3 \subseteq \{0,1\}^{r_3}$ with $\mu(E_3) \leq \varepsilon_3 n$ such that for every $s_3 \in \{0,1\}^{r_3} \setminus E_3$, $v \in [r] \setminus B_t$, and $b \in [k_3]$ it holds that $\widetilde{w}_b^{(v)}(s_3) = w_b^{(v)}$.*

Proof. Fix $v \in [r] \setminus B_t$. By the definition of B_t, one has that $\mathrm{dist}(t^{(v)}, \widetilde{t}^{(v)}) \leq \delta_3$. By the encoding function, $t^{(v)} = C_3(w^{(v)})$. Therefore, for every $b \in [k_3]$,

$$\Pr_{s_3 \sim U_{r_3}} [D_3(b, \widetilde{t}^{(v)}, s_3) \neq w_b^{(v)}] \leq \varepsilon_3.$$

The proof then follows by taking the union bound over all $v \in [r] \setminus B_t$ and $b \in [k_v]$ as indeed $\sum k_v \leq n$. □

For $(u, j) \in [\ell] \times [n_2]$ we define the function $\widetilde{z}_j^{(u)} : \{0,1\}^{r_3} \to \Sigma$ as follows. For $s_3 \in \{0,1\}^{r_3}$ we have $\widetilde{z}_j^{(u)}(s_3) = \widetilde{w}_{j'}^{(v)}(s_3)$, where $(v, j') = \Gamma(u, j) \in [r] \times [k_v]$. Further define the function $\widetilde{z}^{(u)} : \{0,1\}^{r_3} \to \Sigma^{n_2}$ by

$$\widetilde{z}^{(u)}(s_3) = \widetilde{z}_1^{(u)}(s_3) \circ \cdots \circ \widetilde{z}_{n_2}^{(u)}(s_3).$$

Claim 69 (Author Claim 69)**.** *For every $u \notin B_z$ and $s_3 \in \{0,1\}^{r_3} \setminus E_3$ it holds that*

$$\mathrm{dist}(\widetilde{z}^{(u)}(s_3), z^{(u)}) \leq \delta_2.$$

Proof. Fix $s_3 \in \{0,1\}^{r_3} \setminus E_3$ and consider any $u \in [\ell] \setminus B_z$. By the encoding function, for every $j \in [n_2]$ it holds that $z_j^{(u)} = w_{j'}^{(v)}$, where $(v, j') = \Gamma(u, j)$. As $v \notin B_z$, at most $\delta_2 n_2$ of $j \in [n_2]$ satisfy $v \in B_w$. For every other j,

$$\widetilde{z}_j^{(u)} = \widetilde{w}_{j'}^{(v)}(s_3) = w_{j'}^{(v)} = z_j^{(u)},$$

proving the claim. □

For $u \in [\ell]$, $a \in [k_2]$ we define the function $\widetilde{y}_a^{(u)} : \{0,1\}^{r_2} \times \{0,1\}^{r_3} \to \Sigma$ as follows. On $(s_2, s_3) \in \{0,1\}^{r_2} \times \{0,1\}^{r_3}$,

$$\widetilde{y}_a^{(u)}(s_2, s_3) = D_2(a, \widetilde{z}^{(u)}(s_3), s_2).$$

Claim 70 (Author Claim 70)**.** *There exists a set $E_2 \subseteq \{0,1\}^{r_2}$ with $\mu(E_2) \leq \varepsilon_2 n$ such that for every $u \in [\ell] \setminus B_z$, $a \in [k_2]$, and $(s_2, s_3) \in (\{0,1\}^{r_2} \setminus E_2) \times (\{0,1\}^{r_3} \setminus E_3)$ it holds that $\widetilde{y}_a^{(u)}(s_2, s_3) = y_a^{(u)}$.*

Proof. Fix $u \in [\ell] \setminus B_z$. By the encoding function $z^{(u)} = C_2(y^{(u)})$. Recall that

$$\widetilde{y}_a^{(u)}(s_2, s_3) = D_2(a, \widetilde{z}^{(u)}(s_3), s_2).$$

As $s_3 \notin E_3$, $u \notin B_z$, Claim 69 implies $\mathrm{dist}(\widetilde{z}^{(u)}(s_3), z^{(u)}) \leq \delta_2$. Therefore

$$\Pr_{s_2 \sim U_{r_2}}[D_2(a, \widetilde{z}^{(u)}(s_3), s_2) \neq y_a^{(u)}] \leq \varepsilon_2.$$

The proof then follows by taking the union bound over all $a \in [k_2]$ and $u \in [\ell] \setminus B_z$, and noting that $k_2 \ell = n_1 \leq n$. □

Claim 71 (Author Claim 71)**.** *For every $(s_2, s_3) \in (\{0,1\}^{r_2} \setminus E_2) \times (\{0,1\}^{r_3} \setminus E_3)$, it holds that*

$$\mathrm{dist}(\widetilde{y}(s_2, s_3), y) \leq \delta_1,$$

where $\widetilde{y}(s_2, s_3)$ is the concatenation of the k_2-length strings $(\widetilde{y}^{(u)}(s_2, s_3) \mid u \in [\ell])$.

Proof. Note that by Claim 70, the projection of the two strings $\widetilde{y}(s_2, s_3), y$ to a block corresponding to $u \notin B_z$ are in full agreement. The proof then follows by Claim 67. □

We now conclude the proof of Proposition 65. Let $p \in [k]$, by Claim 71, for every $(s_2, s_3) \in (\{0,1\}^{r_2} \setminus E_2) \times (\{0,1\}^{r_3} \setminus E_3)$, we have that $\mathrm{dist}(\widetilde{y}(s_2, s_3), y) \leq \delta_1$. Since by the encoding function $y = C_1(x)$, it holds

$$\Pr_{s_1 \sim U_{r_1}}[D_1(p, \widetilde{y}(s_2, s_3), s_1) \neq x_p] \leq \varepsilon_1.$$

The proof then follows since $\mu(E_2) \leq \varepsilon_2 n$ and $\mu(E_3) \leq \varepsilon_3 n$. □

6.2.1 Proof of Theorem 5

In this short section we prove Theorem 5. We focus on the version that is based on non-explicit samplers, yielding non-explicit reductions. The explicit reduction, which entails a bit more technical work, is deferred to Section 6.3 and Section 6.6. We choose to focus on the non-explicit version first because we believe that understanding LDC in the information-theoretic level is, at present, a deeper and more urgent problem than the question of explicitness. Also, the parameters are easier to work with. For the information-theoretic version, we make use of the sampler that is given by Theorem 15. From here on we refer to the constant $c_{\mathrm{samp}} \geq 1$ that appears in that theorem.

Author proof of Theorem 72, stated at the start of this item. Take C_1 to be the code C in the hypothesis of the theorem, namely, a code with block length $n_1 = n$ and distance $\delta_1 = \delta$. Recall that in the distance amplification procedure from Section 6.1, we make use of a $(\delta_2/2, \delta_1)$ sampler $G = ([\ell], [r], E)$ with $\ell = n_1/k_2$ and left-degree $d = n_2$. For the proof, we will instantiate the distance

amplification procedure with the sampler that is given by Theorem 15. We take C_2 to be an asymptotically good code over Σ set with block length

$$n_2 = c_{\text{samp}} \cdot \frac{\log(1/\delta_1)}{(\delta_2/2)^2} = O(\log(1/\delta)),$$

where δ_2 is the (constant) distance of C_2, having rate at least $1/2$. Note that this choice of parameters is as required by Theorem 15 from the left degree of the sampler. Clearly, C_2 has query complexity $O(\log(1/\delta))$ and error $\varepsilon_2 = 0$. As for the degree D_v of any given right vertex v of the sampler, note that the average right degree is

$$D = \frac{\ell d}{r} = \frac{d}{\delta \log(1/\delta)} = \frac{4c_{\text{samp}}}{\delta_1 \delta_2^2} = \Theta\left(\frac{1}{\delta}\right).$$

Recall that, by Theorem 15, $D_v \in [D/2, 2D]$. For every length in this range, we take a code from the family C' having the required message length. We would like take the family of codes C_3 to be C' though we must reduce the error first. Indeed, note that the error ε of the code obtained by Proposition 65 is $\varepsilon_1 + n(\varepsilon_2 + \varepsilon_3)$. As mentioned in the introduction, one can reduce the error from $1/5$ to $1/(10n)$ by applying the decoding procedure for $c \log n$ times, where c is some large enough constant, and output the symbol according to plurality. This has no effect on the rate or distance of C', and has a multiplicative $O(\log n)$ cost in query complexity. That is, the query complexity of C_3 is $O(q'_{O(1/\delta_1)} \log n)$. The proof then readily follows by Proposition 65. □

6.2.1.1 Improving the query complexity further given low-error LDC

We remark that, if C' has error $O(1/n)$ to begin with, n being the block length of C, then one can skip the error reduction in the proof of Theorem 72, and get a slightly better query complexity. Indeed, this will save the $\log n$ factor in Equation (6.2). Moreover, observe that C_2 can be taken to be an LDC as well, rather than a standard code, which will reduce its deterioration on the query complexity from $O(\log(1/\delta))$ to $q'_{O(\log(1/\delta))}$. However, for that, one need the error of C_2 to be $O(1/n)$ as well. Assuming one can obtain such low-error LDC (note that an error of $1/n$ is at least exponentially-small in the length of C_2 since $\delta > 1/n$), the query complexity can be improved further to

$$q_{\text{new}} \leq q \cdot q'_{O(1/\delta)} q'_{O(\log(1/\delta))}.$$

We conclude this section by instantiating Theorem 72 with C' taken to be the state-of-the-art construction of asymptotically good LDC.

Theorem 73 (Author Theorem 73, [26])**.** *Let Σ be a finite alphabet. Then, there exist constants δ, ρ and an explicit infinite family of $(q_k, \delta, 1/5)$-LDC, k being the message length, having query complexity $q_k = 2^{O(\sqrt{\log(k) \log\log k})}$.*

Using it, one gets query complexity

$$q_{\text{new}} \leq q \log(n) \cdot 2^{O(\sqrt{\log(1/\delta) \cdot \log\log(1/\delta)})} = q \log(n) (1/\delta)^{o(1)}.$$

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